

Project Title
(Pharma Search)

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

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ABSTRACT

Pharma Search is an intelligent medicine search and recommendation system designed to make medicine retrieval faster, more accurate, and easier to use in pharmacies, hospitals, and healthcare organizations. The system allows users to search for medicines using attributes such as medicine name, manufacturer, dosage, and category, while providing relevant medicine details and suggesting suitable substitutes when the requested medicine is unavailable. The project focuses on efficient retrieval from medicine datasets using Data Structures and Algorithms. For exact and efficient pattern searching, the system applies stringmatching algorithms such as Knuth-Morris-Pratt (KMP) and Rabin-Karp. To handle spelling mistakes and approximate medicine names, Edit Distance using Dynamic Programming is used for fuzzy matching and spelling correction. Hashing, arrays, and other suitable data structures are used to organize and retrieve medicine records efficiently. The application is implemented entirely in Java, with medicine records stored and managed using TXT files. By reducing search time, minimizing failed searches caused by spelling errors, and helping users find suitable alternatives, Pharma Search demonstrates the practical application of advanced Data Structures and Algorithms to a real-world healthcare information-retrieval problem.

Signature of the Guide